

ML-TEACHER PRD

Problem 1: Knowledge

Books



Chunking



Embeddings



Vector DB



Retriever

This is solved by RAG.

Problem 2: Teaching Style

Instruction



Definition



Math Intuition



Code Example



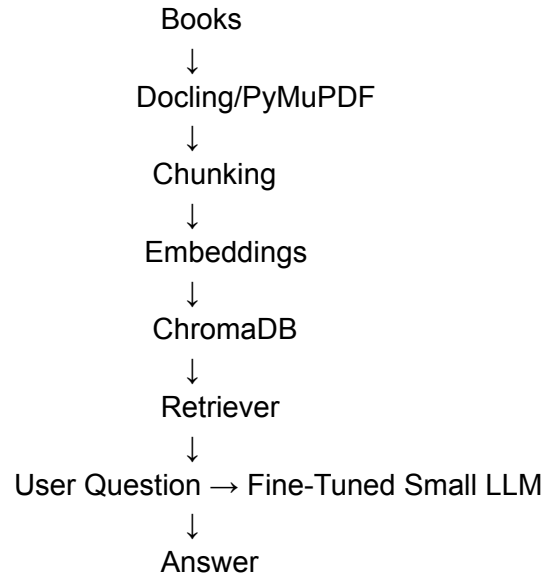
Interview Question



Common Mistakes

This is solved by fine-tuning.

Combined Architecture



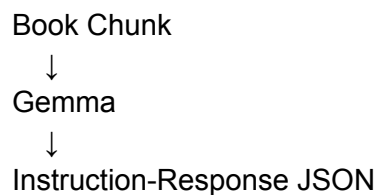
The RAG provides facts.

The fine-tuned model provides the teaching format.

About Gemma Generating the Dataset

You have two options.

Option A: Use Gemma Directly



Example:

Input chunk:

Gradient Descent minimizes a cost function...

Output:

```
{  
  "instruction": "Explain Gradient Descent",  
  "response": "Definition...\nMath Intuition...\nCode Example..."  
}
```

This is the fastest path.

Option B: Use Rules + Gemma

Instead of asking Gemma to invent everything:

Chunk
↓
Extract Topic
↓
Prompt Gemma
↓
Generate Structured Response

This tends to produce cleaner datasets.

For your first version, I'd just use Gemma.

Model Choice

For learning and local experimentation:

Tiny Models

- TinyLlama
- Qwen2.5-0.5B

Good:

Fast training
Low VRAM
Cheap

Bad:

Limited reasoning
Limited coding ability

Better Choice

Since you're teaching ML concepts:

- Qwen2.5-1.5B

This is probably the sweet spot.

Training Dataset Size

For behavior tuning:

100 examples → proof of concept
500 examples → noticeable improvement
1000+ examples → good consistency

You do not need 50,000 examples.

You are teaching format, not world knowledge.

What the Fine-Tuned Model Learns

Not:

Gradient Descent facts
Random Forest facts
PCA facts

RAG handles those.

It learns:

How to explain
How to structure
How to teach
How to present code

This is exactly the kind of task LoRA excels at.

Stage 1

Book
↓
Chunk
↓
Embedding
↓
ChromaDB
↓
RAG

Get retrieval working.

Stage 2

Book
↓
Gemma
↓
500–1000 JSON pairs
↓
LoRA Fine-Tune

Get the teaching style working.

Stage 3

RAG
+
Fine-Tuned Model

Now you have both knowledge and teaching behavior.

Don't use a *too tiny* model just because it's small. **Qwen 2.5 1.5B** or **3B** for the tutor model. The extra quality is often worth it, especially for ML concepts where clarity matters.